EARTH SYSTEM COMPONENTS

12-GEO-M-AG02

Lectures: Miguel D. Mahecha with contributions by Ana Bastos

Exercises: Milena Mönks

Remote Sensing Centre for Earth System Research - rsc4earth.de



UNIVERSITÄT
LEIPZIG

Before Christmas: the atmosphere first, then water

Block	L	Date	Topic
Intro	L01	13.10.2026	Course overview and organisation
I · Atmosphere	**L02**	**20.10.2026**	**Atmosphere I: concepts and key variables (today)**
	L03	27.10.2026	Atmosphere II: radiation balance
	L04	03.11.2026	Atmosphere III: structure of the atmosphere
	L05	10.11.2026	Atmosphere IV: buoyancy forces
	L06	17.11.2026	Atmosphere V: circulation
II · Terrestrial hydrology	L07	24.11.2026	Terrestrial hydrology I: water-cycle concepts
	L08	01.12.2026	Terrestrial hydrology II
	L09	08.12.2026	Terrestrial hydrology II: measurements, Budyko framework
III · Biogeochemistry	L10	15.12.2026	Biogeochemistry I: element cycles – the carbon cycle

After Christmas: carbon, classification and special lectures

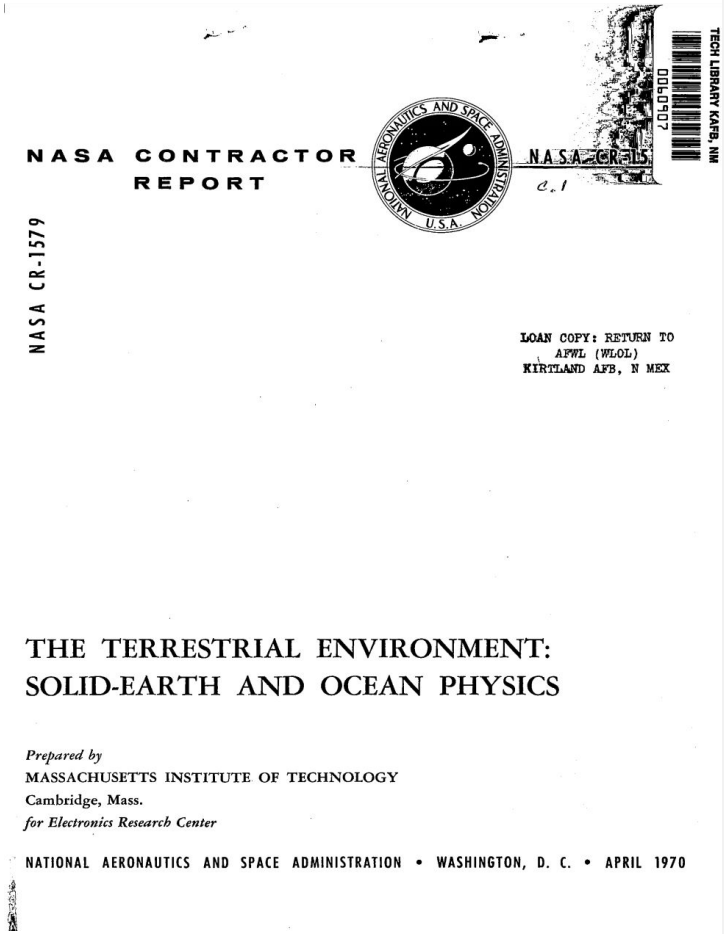
Block	L	Date	Topic
–	–	*22.12.–02.01.*	*Christmas break*
III · Biogeochemistry	L11	05.01.2027	Biogeochemistry II: terrestrial biogeochemistry
IV · Classification & special topics	L12	12.01.2027	Climate classification systems and their limitations
	L13	19.01.2027	Atmospheric rivers
	L14	26.01.2027	Forest damage and climate change – guest lecture: Prof. Henrik Hartmann
	L15	02.02.2027	Climate classifications and vegetation
–	–	*Feb 2027*	*Oral exams (dates tbc)*

RECAP LAST WEEK

What did stick to your mind?

- …

Forgot to share this



NASA CONTRACTOR REPORT

NASA CR-1579

NASA CR-1579

LOAN COPY: RETURN TO
AFWL (WLOL)
KIRTLAND AFB, N MEX

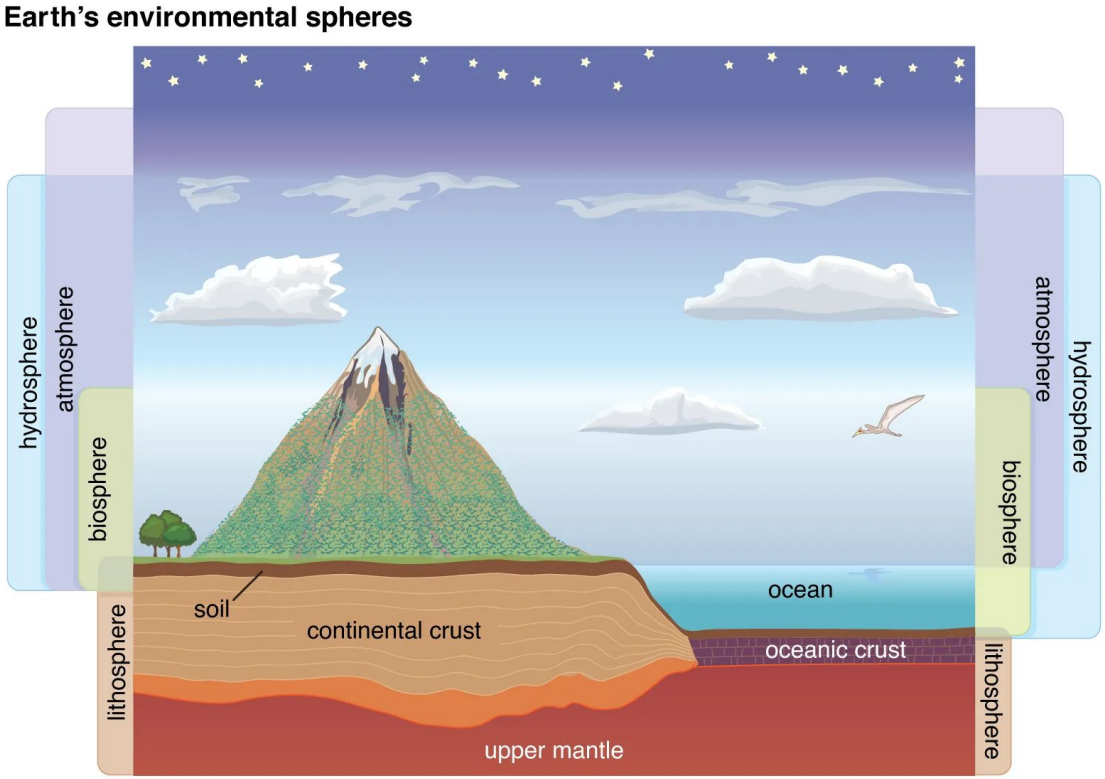
THE TERRESTRIAL ENVIRONMENT:
SOLID-EARTH AND OCEAN PHYSICS

Prepared by
MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Cambridge, Mass.
for Electronics Research Center

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION • WASHINGTON, D. C. • APRIL 1970

https://ilrs.gsfc.nasa.gov/docs/williamstown_1968.pdf

How do these spheres interact? Your solutions please!



© Encyclopædia Britannica, Inc.

How do these spheres interact?

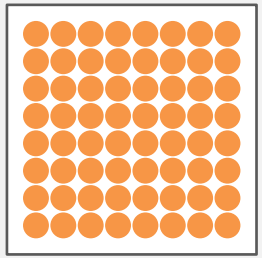
What is the (fraction of) mass of each sphere, in which the main interactions in contemporaneous Earth system dynamics takes place?

What is the (fraction of) volume of each sphere, in which the main interactions in contemporaneous Earth system dynamics takes place?

I. ATMOSPHERE

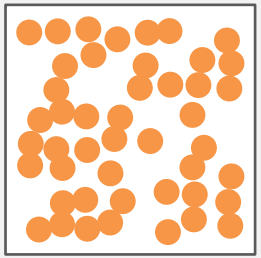
Before we start: three states of matter

The state of matter of a substance depends on temperature and pressure.



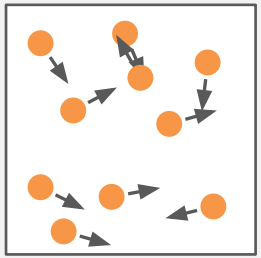
Solid

- **Shape and volume are kept.** Hardly compressible.
- Particles are packed closely and regularly and vibrate around fixed positions.



Liquid

- **Volume is kept, shape adapts.** Hardly compressible.
- Particles are packed closely but can move past each other.



Gaseous

- **Gases fill any available space.** Easily compressible.
- Particles are far apart and move freely.

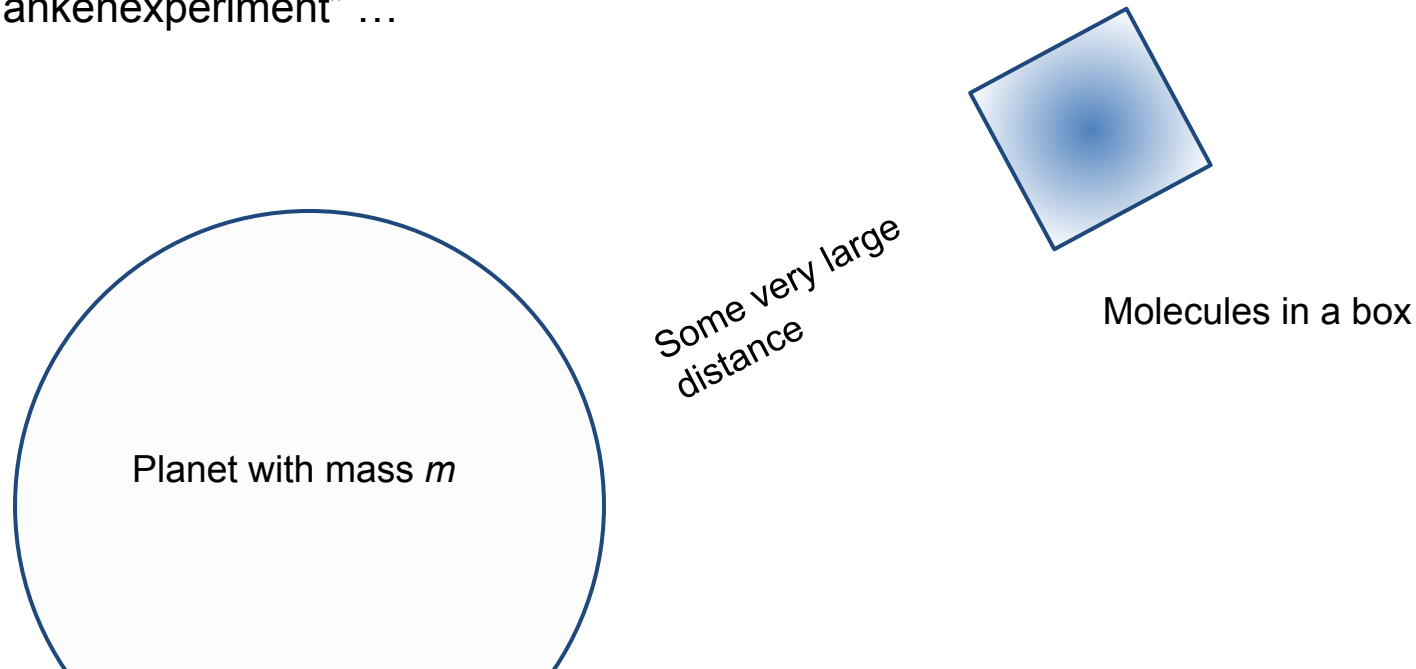
What is an atmosphere?

“ἀτμός” (*atmós*) “Vapour”, “Steam”, “σφαῖρα” (*sfaira*) “Sphere”

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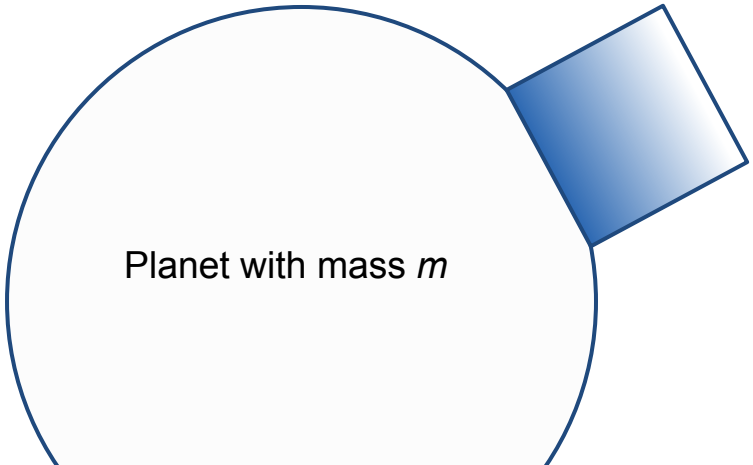
- A "Gedankenexperiment" …



What is an atmosphere?

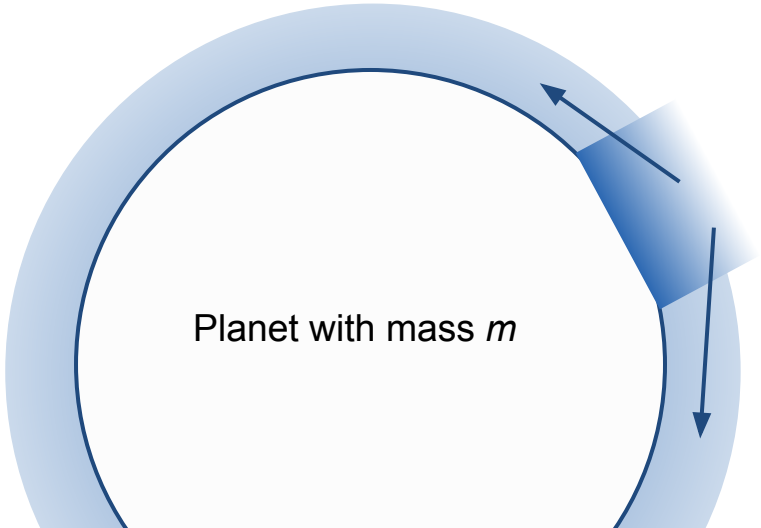
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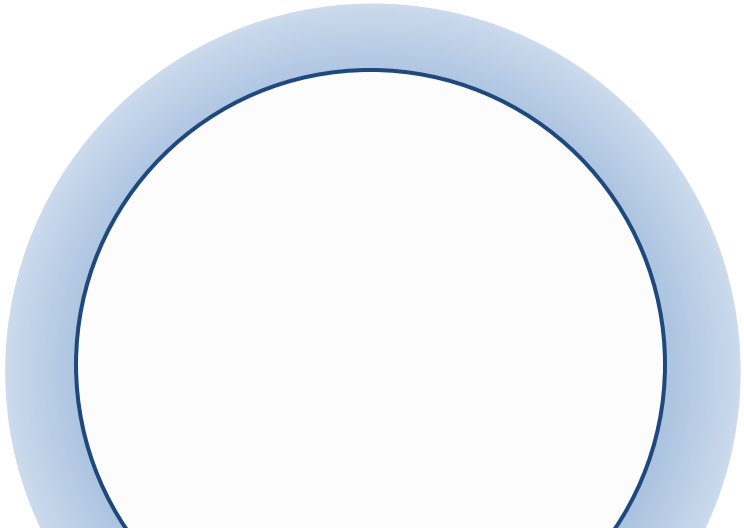
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What is an atmosphere?

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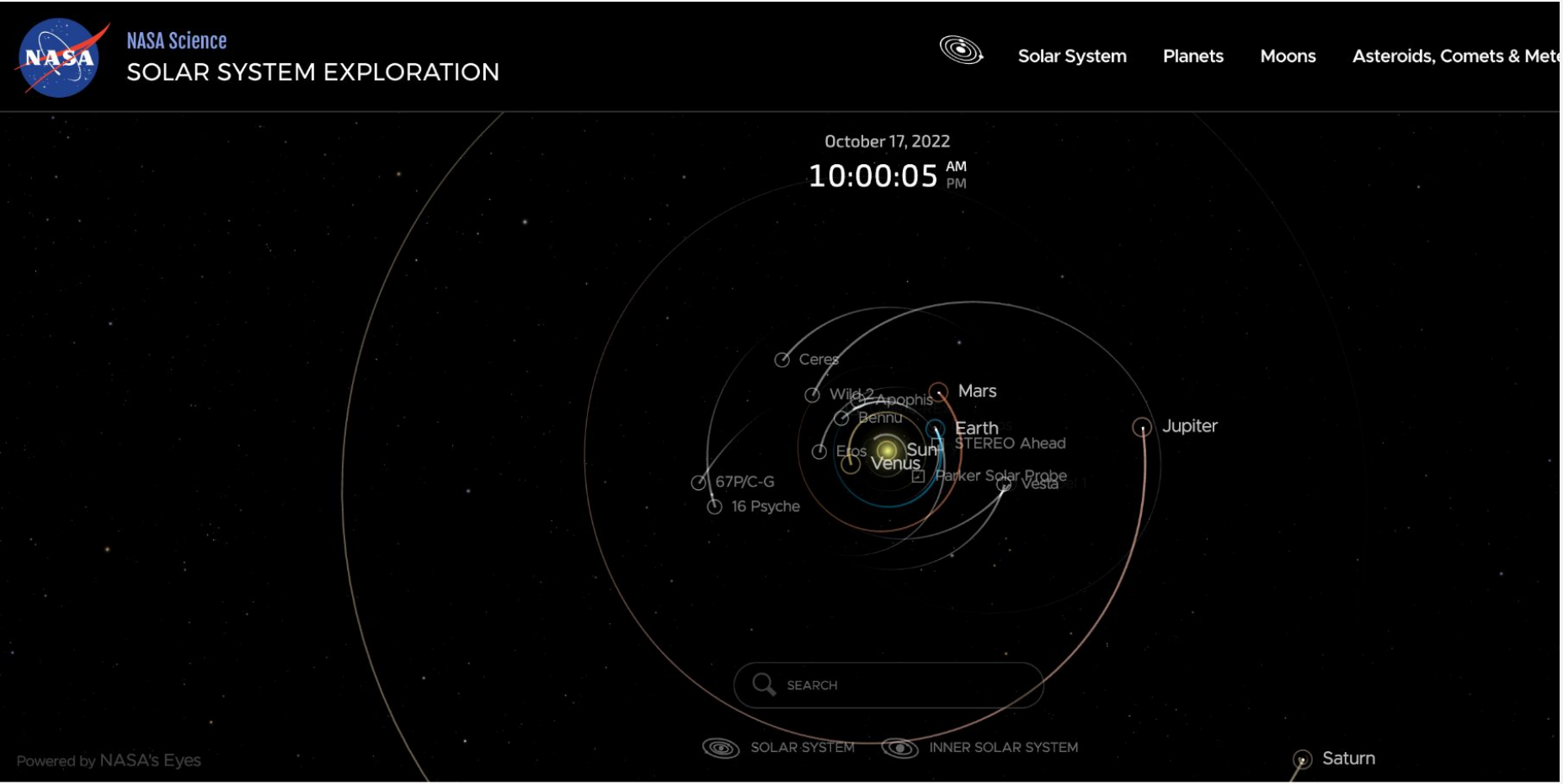
- Proportionen are rubbish
- According to the Fédération aéronautique internationale (FAI), the Earth's atmosphere and outer space, are divided by the “Kármán-Line” at 100 km



$$\frac{z_\mathrm{K}}{r_\mathrm{E}} = \frac{100\,[\mathrm{km}]}{6371\,[\mathrm{km}]} \approx 0.016 \approx 1.6\,\%$$

z_K Kármán line (FAI convention), 100 [km]

r_E mean radius of the Earth, 6371 [km]



https://solarsystem.nasa.gov/planets/overview/

A short note on units

- We describe the Earth system with physical quantities, i.e. “numbers” that measure a physical phenomenon (“length”, “time”, etc.).
- No measurement without a unit (“length in metres”, “time in seconds”, etc.).
- Treat a quantity as the product of a numerical value and a unit …
- Units as a description: the speed of a body, for example, is given as a quotient, e.g. in m/s
- A few base units suffice to describe most physical processes
- The “Système international d’unités” (SI) is the internationally agreed system of units.

Important SI (Système international d'unités) base quantities for us

Quantity	Symbol	Unit, unit symbol
Length	l	Metre, m
Time	t	Second, s
Mass	m	Kilogram, kg
Temperature	T	Kelvin, K
Amount of substance	n	Mole, mol

Units are "carried along" in calculations. Example: the distance a bicycle travels at constant speed in a given time is speed times time, so the unit is m/s · s = m. Further SI base units: ampere (A, electric current) and candela (cd, luminous intensity) – we do not need them here.

Derived units we use often

Quantity	Unit	SI base units	Example in the lecture
Force	Newton, N	kg m s^{-2}	Weight (L05)
Pressure	Pascal, Pa = N m^{-2}	kg m^{-1} s^{-2}	Air pressure 1013 hPa
Energy	Joule, J = N m	kg m^{2} s^{-2}	R = 8.314 J mol^{-1} K^{-1}
Power	Watt, W = J s^{-1}	kg m^{2} s^{-3}	Solar constant 1361 W m^{-2}
Frequency	Hertz, Hz	s^{-1}	Radiation (L03)

Common in meteorology and hydrology:
1 hPa = 100 Pa; 1 bar = 1000 hPa
T [K] = T [°C] + 273.15
1 ppm = 10^{-6}; precipitation: 1 mm = 1 L m^{-2}

How do planets get their atmosphere?

Theory 1: “Primordial atmosphere” (primary atmosphere)

- Forms with the condensing planet, i.e. it is present from the start (for the Earth: hydrogen, helium, methane, ammonia …)
- May be lost over time

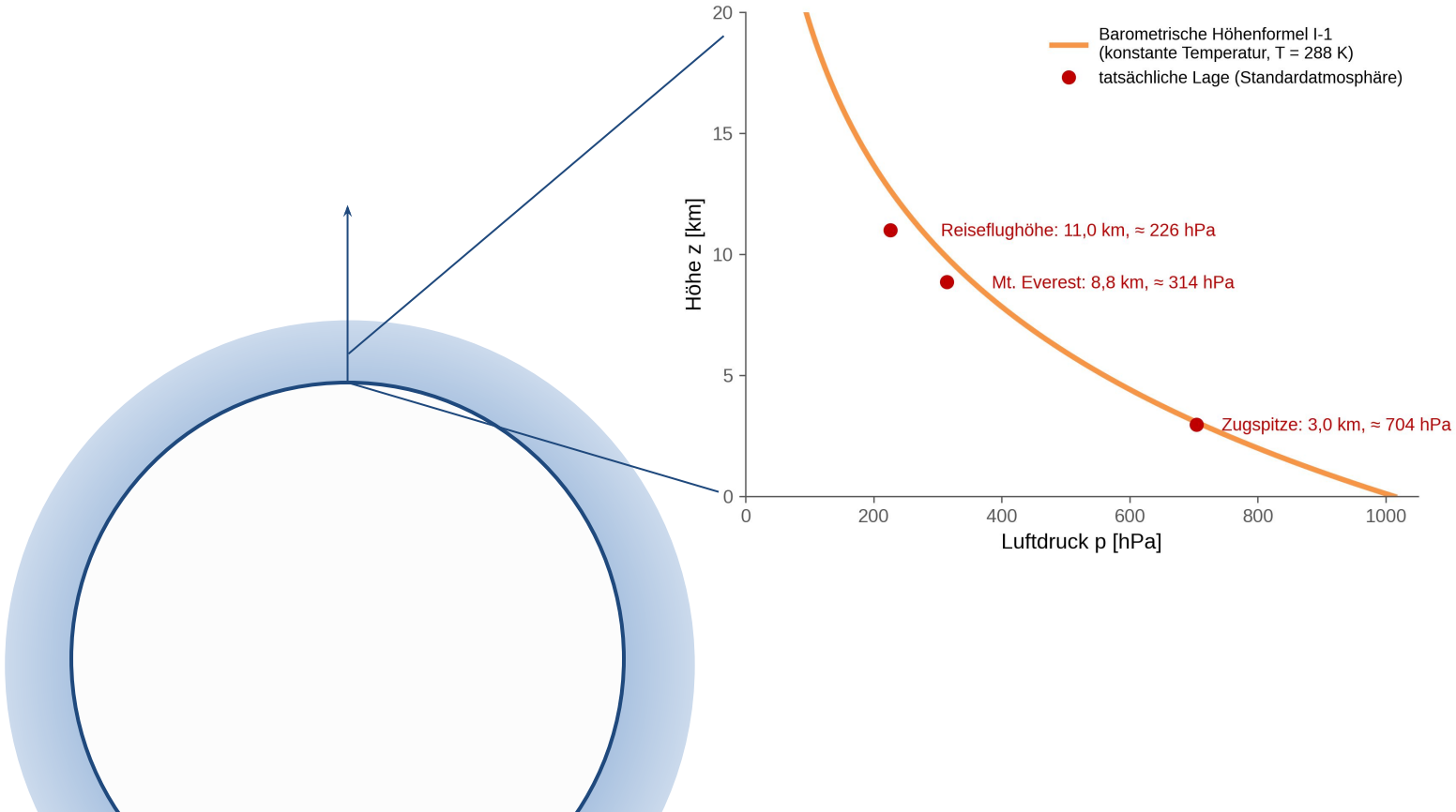
Theory 2: “Outgassed atmosphere”

- Gas mixture supplied from the planet’s interior

Prerequisite

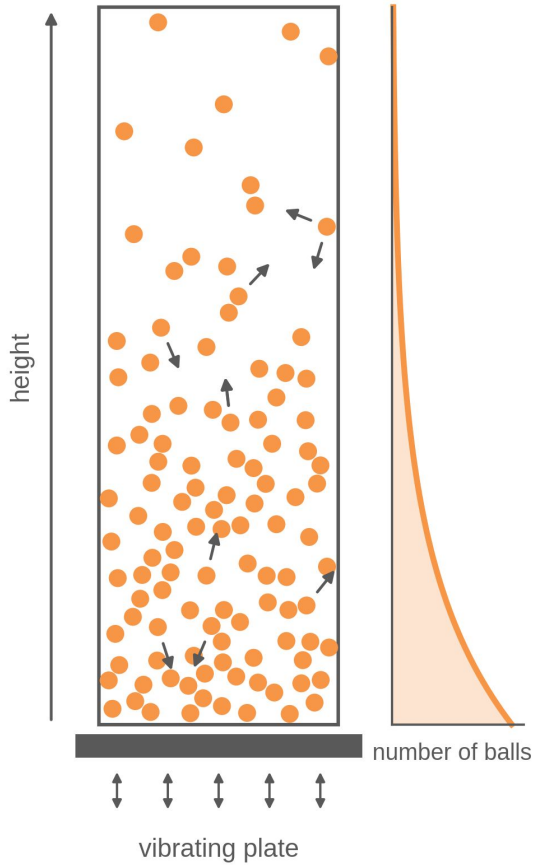
- The planet needs a mass which – together with its radius – determines the gravity field at the surface

Pressure in the atmosphere decreases with height …



Pressure in the atmosphere decreases with height …

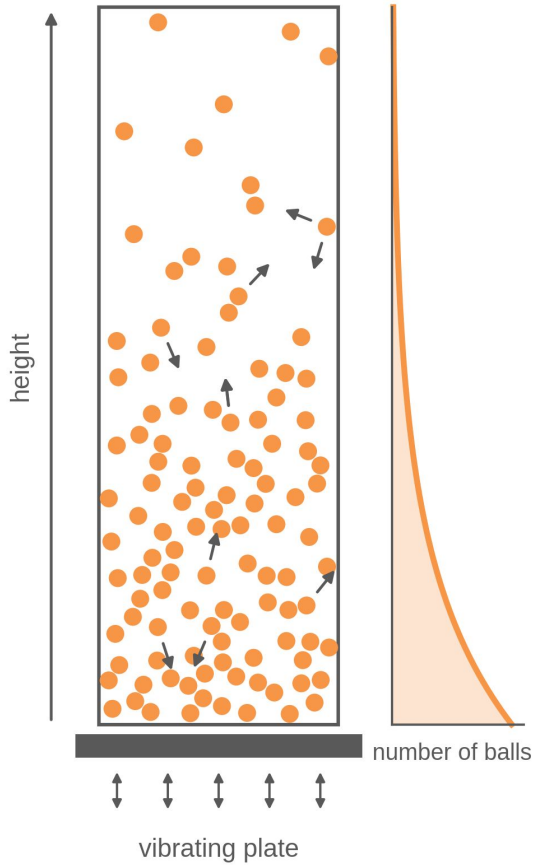
Another thought experiment:
Balls on a vibrating plate

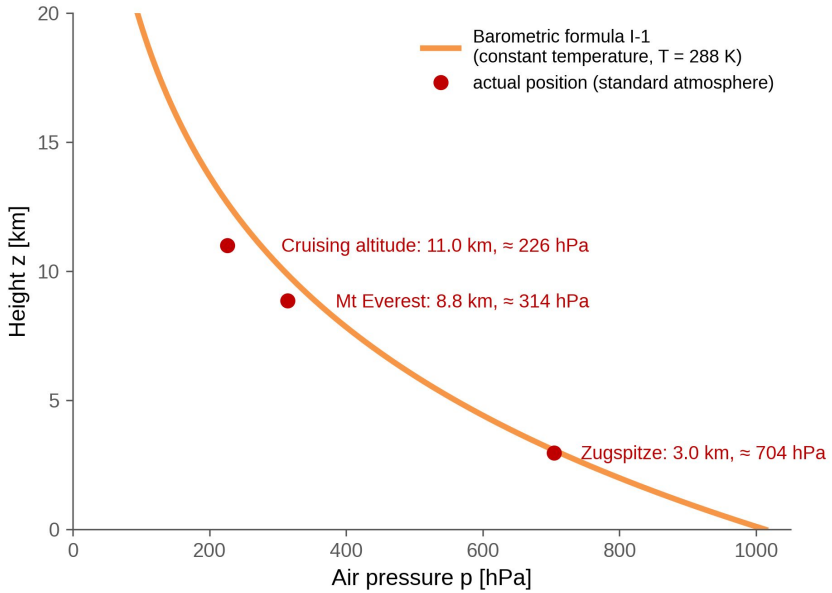


Note: schematic figures without a source were created with AI (Claude Opus 5.5, 2026) and adapted where necessary.

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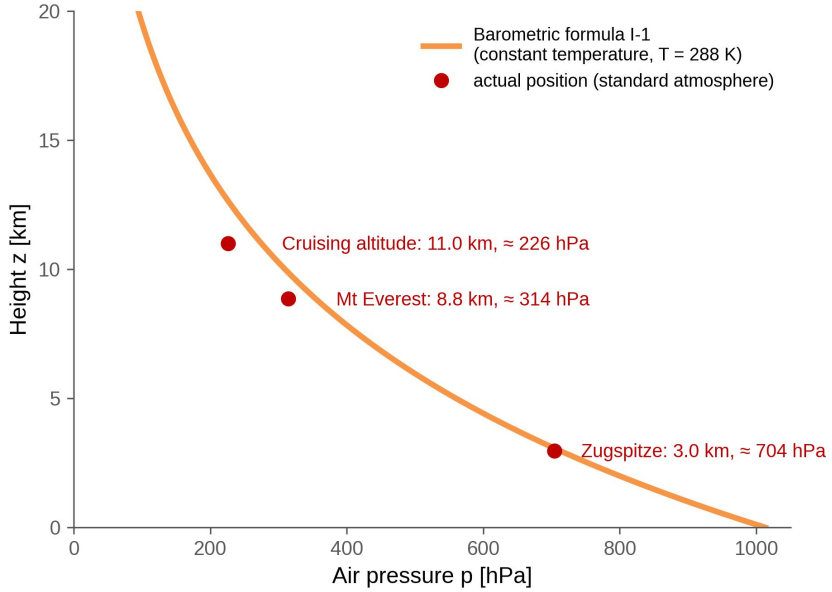
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Pressure in the atmosphere decreases with height …

Formula sheet I-1

$$p(z) = p(z_0)\,\exp\left[-\frac{M_\mathrm{L}\,g}{R\,T}\,(z - z_0)\right]$$

$p(z)$, $p(z_0)$	air pressure at height z and at reference height z_0 [Pa]
z, z_0	height, reference height [m]
M_L	molar mass of dry air, $0.028\,97\,[\mathrm{kg\,mol^{-1}}]$
g	gravitational acceleration at the Earth's surface, $9.81\,[\mathrm{m\,s^{-2}}]$
R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$
T	temperature, assumed constant [K]



The barometric formula describes the decrease of pressure with height in an atmosphere … any problem with the formula?

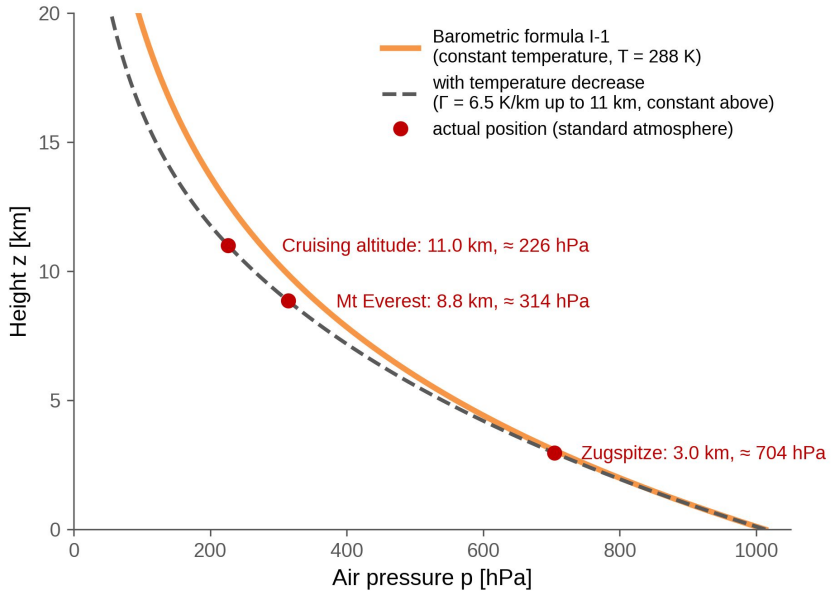
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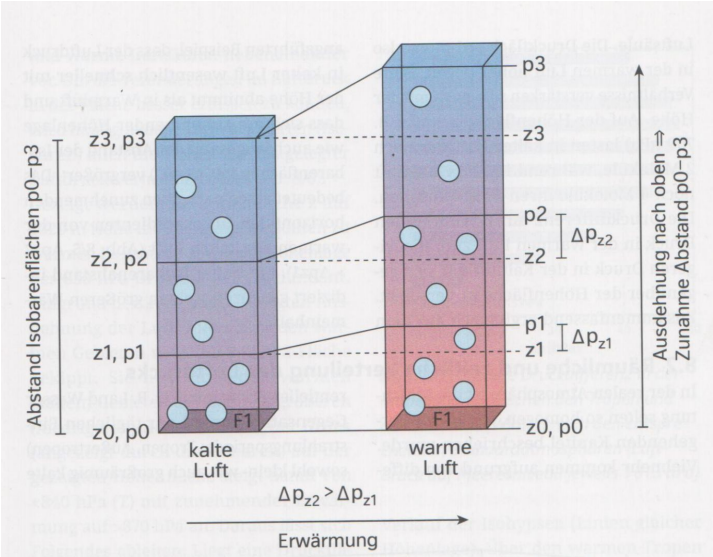
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Assumptions: hydrostatic equilibrium, constant temperature, constant g and M_L.

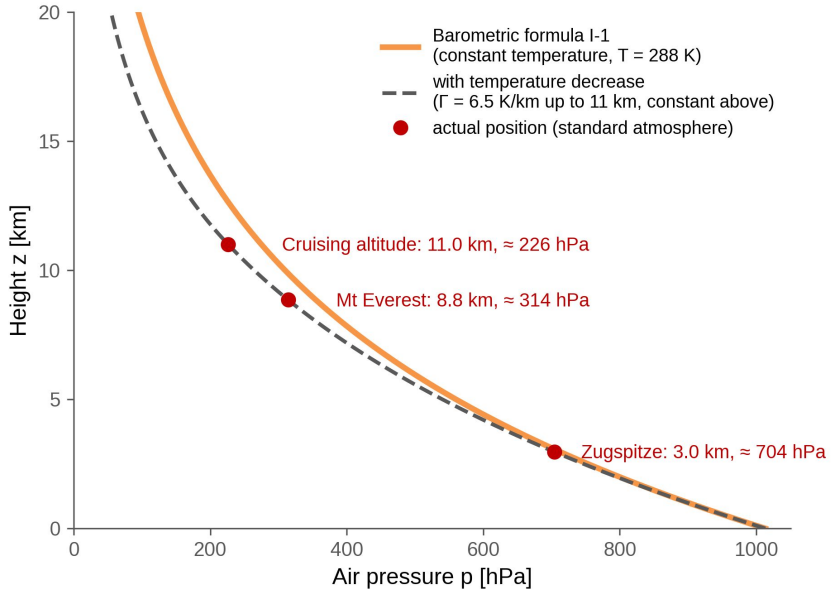


The barometric formula describes the decrease of pressure with height in an atmosphere (assuming constant temperature)

Pressure in the atmosphere decreases with height …

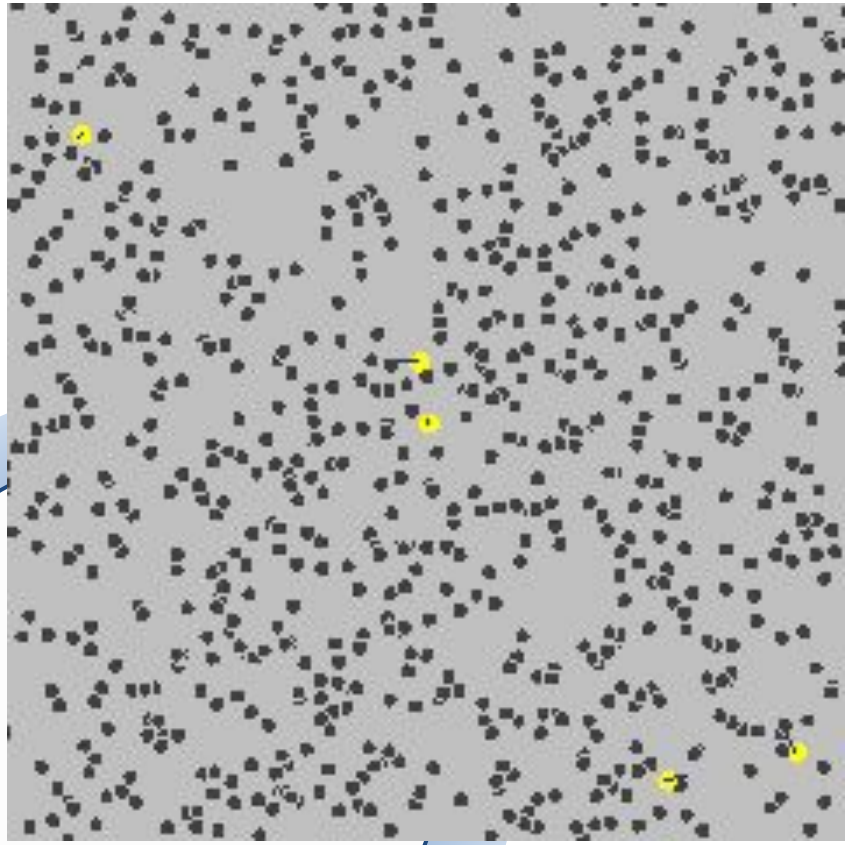


Bendix & Luterbacher (2019)
Klimatologie. Westermann.



The barometric formula describes the decrease of pressure with height in an atmosphere (assuming constant temperature)

Why does a planet retain its atmosphere?



https://en.wikipedia.org/wiki/Brownian_motion

Why does a planet retain its atmosphere?

The escape velocity v_{esc} is the speed needed to escape the gravity field of a planet.

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Planet-specific quantity; for the Earth:

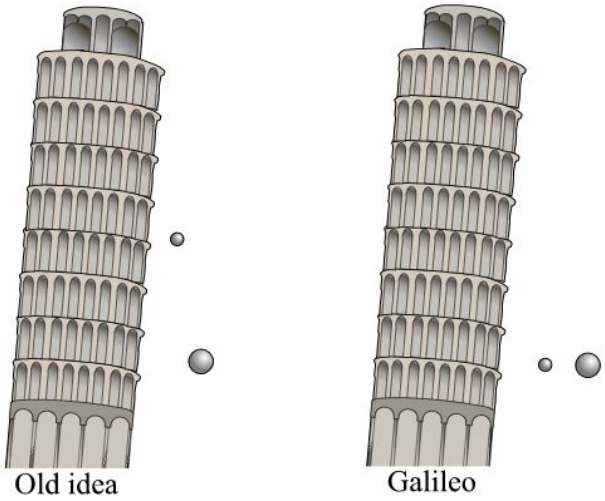
$$v_{\mathrm{esc}} = \sqrt{2\, g\, r_{\mathrm{E}}}$$

Formula sheet I-**2**

v_{esc}	escape velocity $[\mathrm{m\,s^{-1}}]$
g	gravitational acceleration at the Earth's surface, $9.81\,[\mathrm{m\,s^{-2}}]$
r_{E}	mean radius of the Earth, $6.371 \times 10^{6}\,[\mathrm{m}]$

Spherical Earth, no air resistance, no rotation. For other celestial bodies: g_{K} and r_{K} (formula sheet I-2).

A small anecdote:





David Scott does the feather hammer experiment on the moon

Why does a planet retain its atmosphere?

The escape velocity v_{esc} is the speed needed to escape the gravity field of a planet.

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Spherical Earth, no air resistance, no rotation. For other celestial bodies: g_K and r_K (formula sheet I-2).

v_{esc} therefore depends on r and g and is specific to each planet – but independent of the size and mass of the object! (Whether molecule or rocket.)

Why does a planet retain its atmosphere?

The escape velocity v_{esc} is the speed needed to escape the gravity field of a planet.

Let's do the math for planet Earth:

Formula sheet I-**2**

$$
\begin{aligned}
v_{\mathrm{esc}} &= \sqrt{2\,g\,r_{\mathrm{E}}} \\
&= \sqrt{2 \cdot 9.81\,[\mathrm{m\,s^{-2}}] \cdot 6.371 \times 10^{6}\,[\mathrm{m}]} \\
&\approx 11\,180\,[\mathrm{m\,s^{-1}}] \approx 11.2\,[\mathrm{km\,s^{-1}}]
\end{aligned}
$$

g	gravitational acceleration at the Earth's surface, $9.81\,[\mathrm{m\,s^{-2}}]$
r_{E}	mean radius of the Earth, $6.371 \times 10^{6}\,[\mathrm{m}]$

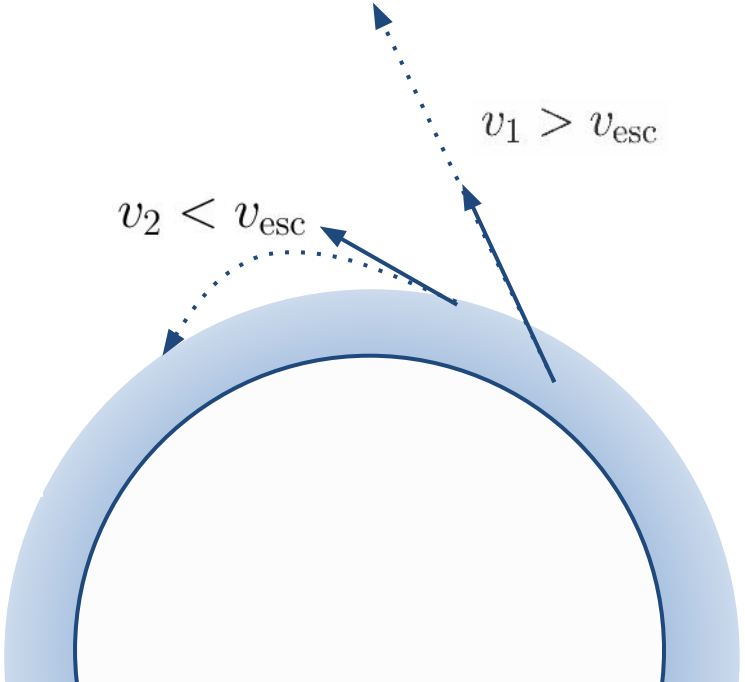
Why does a planet retain its atmosphere?

The escape velocity v_{esc} is the speed needed to escape the gravity field of a planet.

$$v_1 > v_{esc} \quad \Rightarrow \quad \text{molecule escapes}$$
$$v_2 < v_{esc} \quad \Rightarrow \quad \text{molecule falls back}$$
$$v_{esc} \gg v_{rms} \quad \Rightarrow \quad \text{gas is retained in the long run}$$

v_1, v_2	speed of molecule 1 and 2 $[\mathrm{m\,s^{-1}}]$
v_{esc}	escape velocity $[\mathrm{m\,s^{-1}}]$
v_{rms}	root-mean-square molecular speed $[\mathrm{m\,s^{-1}}]$

But how fast are molecules actually?



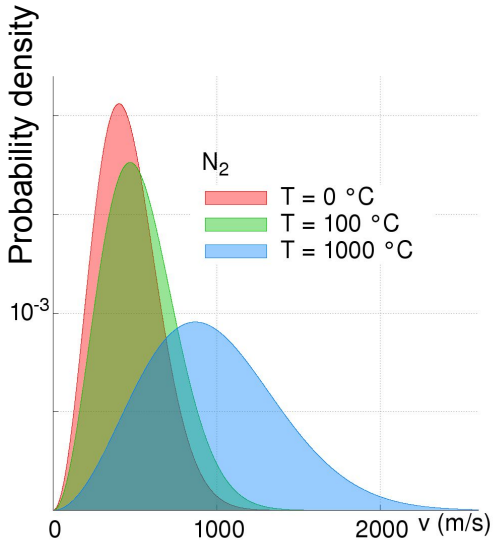
Speed of a molecule

Molecular speeds are distributed (Maxwell–Boltzmann). We calculate with the root-mean-square speed; it depends on the molar mass and the temperature.

$$v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}$$

v_{rms}	root-mean-square molecular speed $[\mathrm{m\,s^{-1}}]$
R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$
T	temperature $[\mathrm{K}]$ $\quad(0\,\mathrm{K} = -273.15\,^{\circ}\mathrm{C})$
M	molar mass of the gas $[\mathrm{kg\,mol^{-1}}]$ $\quad$(e.g. N_2: $0.028\,[\mathrm{kg\,mol^{-1}}]$)

v_{rms} is slightly larger than the most probable speed (maximum of the Maxwell–Boltzmann distribution).



Maxwell–Boltzmann distribution for N2 at different temperatures; Wikipedia (accessed Oct. 2021)

Speed of a molecule

$$v_{\mathrm{rms}} = \sqrt{\frac{3\,RT}{M}}$$

Formula sheet I-**3**

v_{rms}	root-mean-square molecular speed $[\mathrm{m\,s^{-1}}]$
R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$
T	temperature, $288.15\,[\mathrm{K}]$ $(15\,^{\circ}\mathrm{C})$
M	molar mass of the gas $[\mathrm{kg\,mol^{-1}}]$; divide table values in $\mathrm{g\,mol^{-1}}$ by 1000 (N_2: $28\,\mathrm{g\,mol^{-1}} = 0.028\,[\mathrm{kg\,mol^{-1}}]$)
$[\mathrm{J}]$	Joule, $1\,[\mathrm{J}] = 1\,[\mathrm{kg\,m^2\,s^{-2}}]$

$\mathrm{mol^{-1}}$, K *and* kg *cancel; in the last line* M *is the numerical value in* $\mathrm{kg\,mol^{-1}}$. N_2*:* $M = 0.028$ *gives* $v_{\mathrm{rms}} \approx 507\,[\mathrm{m\,s^{-1}}]$.

Whatever happens ...
Mind the units!

Speed of a molecule

$$
\begin{aligned}
v_{\mathrm{rms}} &= \sqrt{\frac{3\,RT}{M}} \\
&= \sqrt{\frac{3 \cdot 8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}] \cdot 288.15\,[\mathrm{K}]}{M\,[\mathrm{kg\,mol^{-1}}]}} \\
&= \sqrt{\frac{3 \cdot 8.314\,[\mathrm{J}] \cdot 288.15}{M\,[\mathrm{kg}]}} \\
&= \sqrt{\frac{3 \cdot 8.314\,[\mathrm{kg\,m^2\,s^{-2}}] \cdot 288.15}{M\,[\mathrm{kg}]}} \\
&= \sqrt{\frac{3 \cdot 8.314\,[\mathrm{m^2\,s^{-2}}] \cdot 288.15}{M}}
\end{aligned}
$$

v_{rms}	root-mean-square molecular speed $[\mathrm{m\,s^{-1}}]$
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Speed of a molecule

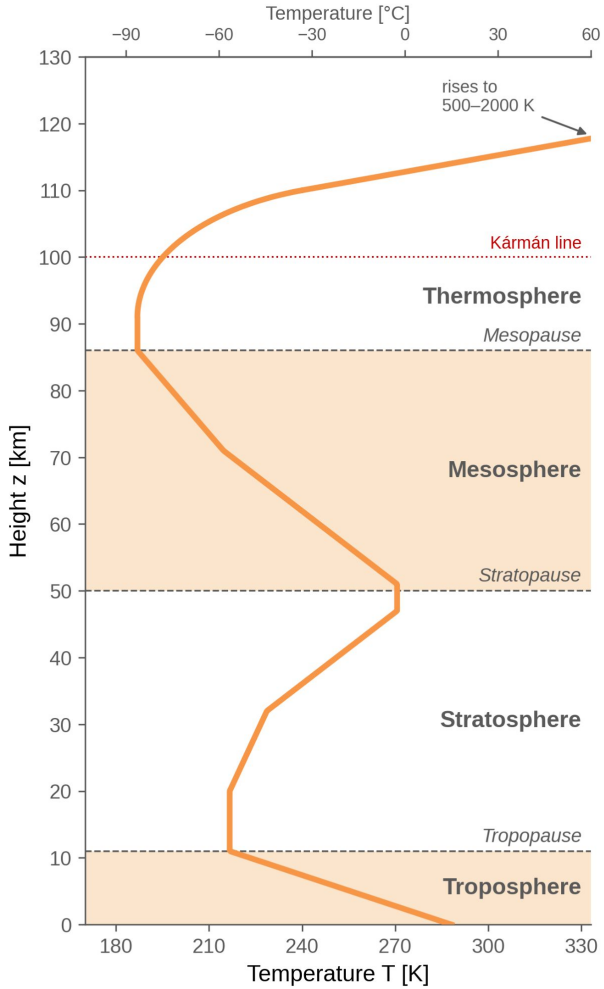
The warmer, the faster the molecules move. Temperatures in the thermosphere can exceed 1000 °C

…

$$v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}$$

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R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$
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v_{rms} is slightly larger than the most probable speed (maximum of the Maxwell–Boltzmann distribution)



Speed of a molecule

$$v_{\mathrm{rms}} = \sqrt{\frac{3\,R\,T}{M}}$$

Substance	Composition	M [g mol^{-1}]	M [kg mol^{-1}]
H	1×1	1	0.001
H_2	2×1	2	0.002
H_2O	$2 \times 1 + 1 \times 16$	18	0.018
N_2	2×14	28	0.028
O_2	2×16	32	0.032
CO_2	$1 \times 12 + 2 \times 16$	44	0.044

M	molar mass (rounded); table value in [g mol^{-1}], insert into formula I-3 in [kg mol^{-1}]
v_{rms}	root-mean-square molecular speed from I-3 [m s^{-1}]
T	temperature, 288.15 [K]

Speed of nitrogen

$$v_{\mathrm{rms}} = \sqrt{\frac{3\,RT}{M}}$$

$$= \sqrt{\frac{3 \cdot 8.314\,[\mathrm{kg\,m^2\,s^{-2}\,mol^{-1}\,K^{-1}]} \cdot 288.15\,[\mathrm{K}]}{0.028\,[\mathrm{kg\,mol^{-1}}]}}$$

$$= \sqrt{2.567 \times 10^5\,[\mathrm{m^2\,s^{-2}}]} \approx 507\,[\mathrm{m\,s^{-1}}]$$

R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$, with $1\,[\mathrm{J}] = 1\,[\mathrm{kg\,m^2\,s^{-2}}]$
T	temperature, $288.15\,[\mathrm{K}]$
M	molar mass of N_2, $0.028\,[\mathrm{kg\,mol^{-1}}]$ $\quad (= 28\,\mathrm{g\,mol^{-1}})$

Always insert the molar mass in $\mathrm{kg\,mol^{-1}}$*. Using the table value in* $\mathrm{g\,mol^{-1}}$ *makes the result too small by a factor of* $\sqrt{1000} \approx 32$*.*

Speed of a molecule

$$v_{\mathrm{rms}} = \sqrt{\frac{3\,R\,T}{M}}$$

Substance	Composition	M [g mol^{-1}]	M [kg mol^{-1}]	v_{rms} [m s^{-1}]
H	1×1	1	0.001	2681
H_2	2×1	2	0.002	1896
H_2O	$2 \times 1 + 1 \times 16$	18	0.018	632
N_2	2×14	28	0.028	507
O_2	2×16	32	0.032	474
CO_2	$1 \times 12 + 2 \times 16$	44	0.044	404

M	molar mass (rounded); table value in [g mol^{-1}], insert into formula I-3 in [kg mol^{-1}]
v_{rms}	root-mean-square molecular speed from I-3 [m s^{-1}]
T	temperature, 288.15 [K]

Which gases remain on Earth (to a first approximation)?

If escape velocity and mean molecular speed were equal, what would happen to the atmosphere?

Which gases remain on Earth (to a first approximation)?

If the molecular speed were similar to the escape velocity v_{esc}, most molecules would leave the gravity field of the planet!

Only if $v_{\mathrm{esc}} \gg v_{\mathrm{mol}}$ molecule cannot leave the atmosphere. The “much greater than” is needed because the speeds are distributed. For long periods, assume a factor of at least 10.

H_2: 1896 · 10 [m/s] > 11180 [m/s] is hard to retain

N_2: 507 · 10 [m/s] < 11180 [m/s] is easily retained

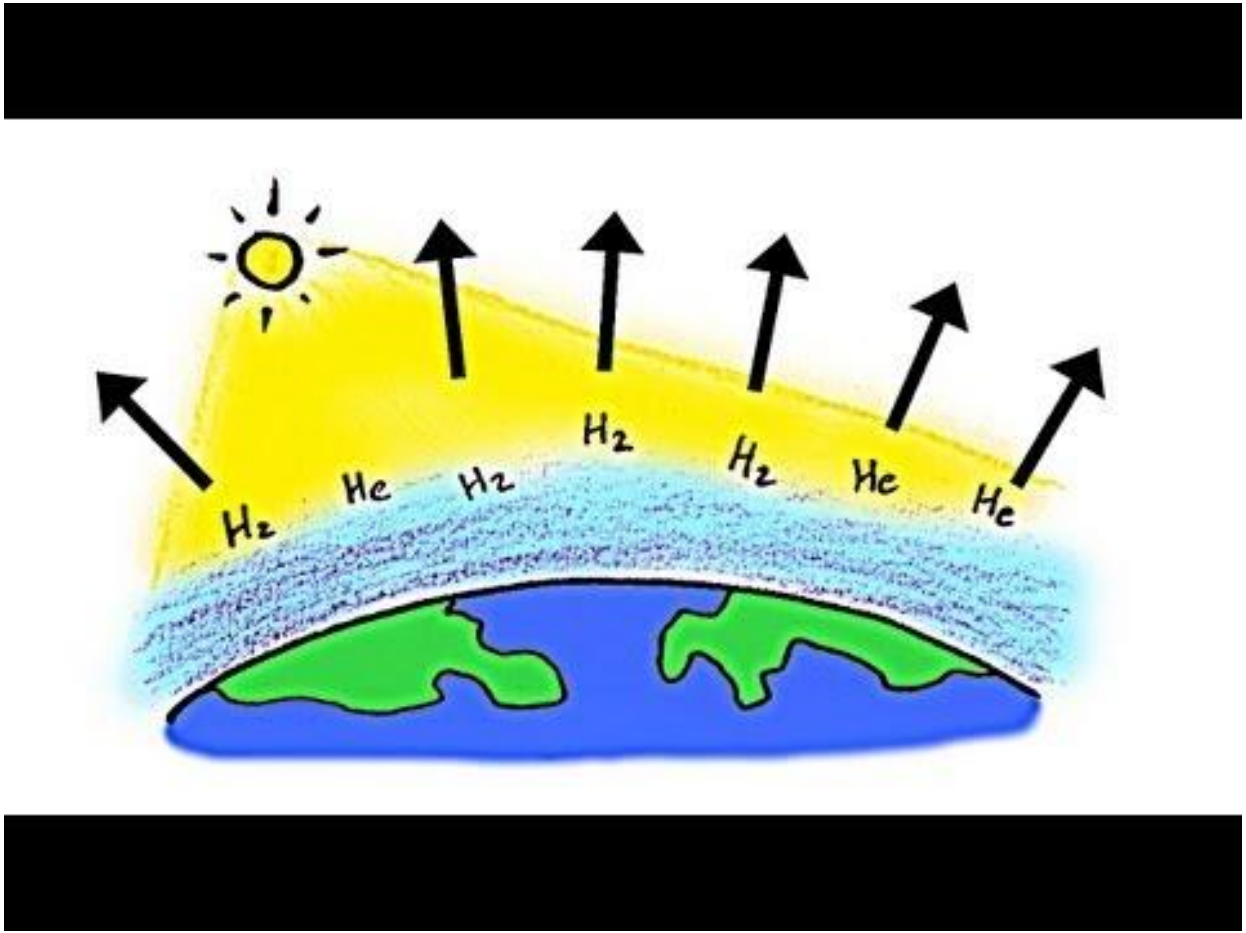
End of thought experiment!
Physics incomplete!

Which gases remain on Earth (to a first approximation)?

"The existence of the Earth's atmosphere rests on the balance between the Earth's gravitational acceleration ('attraction') and the intrinsic motion of the gases that make up the atmosphere. If the motion of the gases were greater than the gravitational acceleration, i.e. greater than the speed needed to leave the Earth's sphere of attraction (namely 11 [km/s]!), the gaseous envelope would escape into space." (translated)

Kuttler (2013) Klimatologie, 2nd ed. https://www.utb.de/doi/book/10.36198/9783838555041

- The planet must have a minimum size (gravity!)
- Temperature must not be too high (the warmer, the faster the molecules)
- Gases must not be too light



MinuteEarth (2013): "Our Atmosphere is Escaping!"
https://www.youtube.com/watch?v=e7ZqMTBwFVs

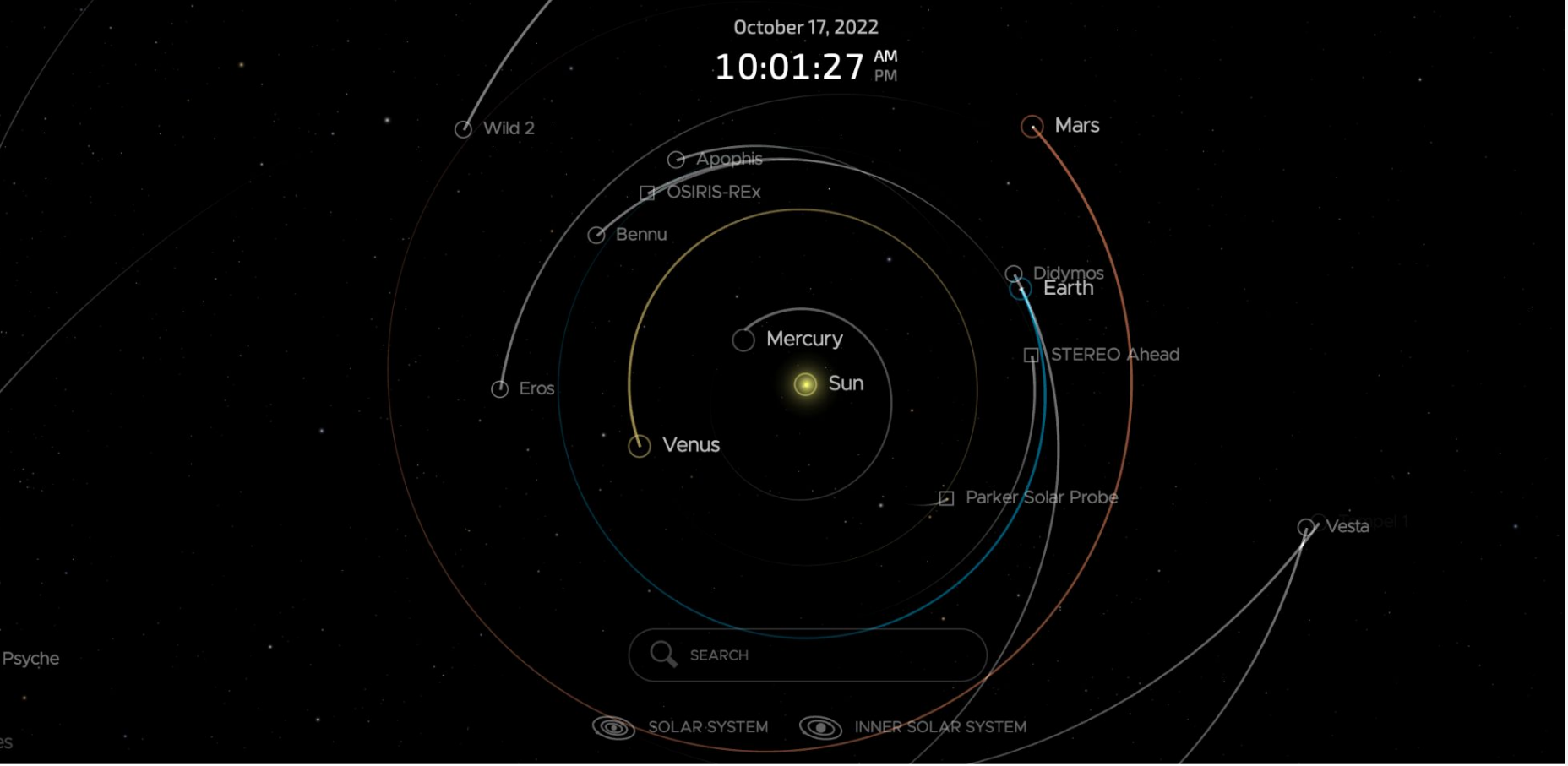
So what do we know about other planets?

Escape velocity depends on g (and r)

The speed of gas molecules depends on the gas and on the temperature T

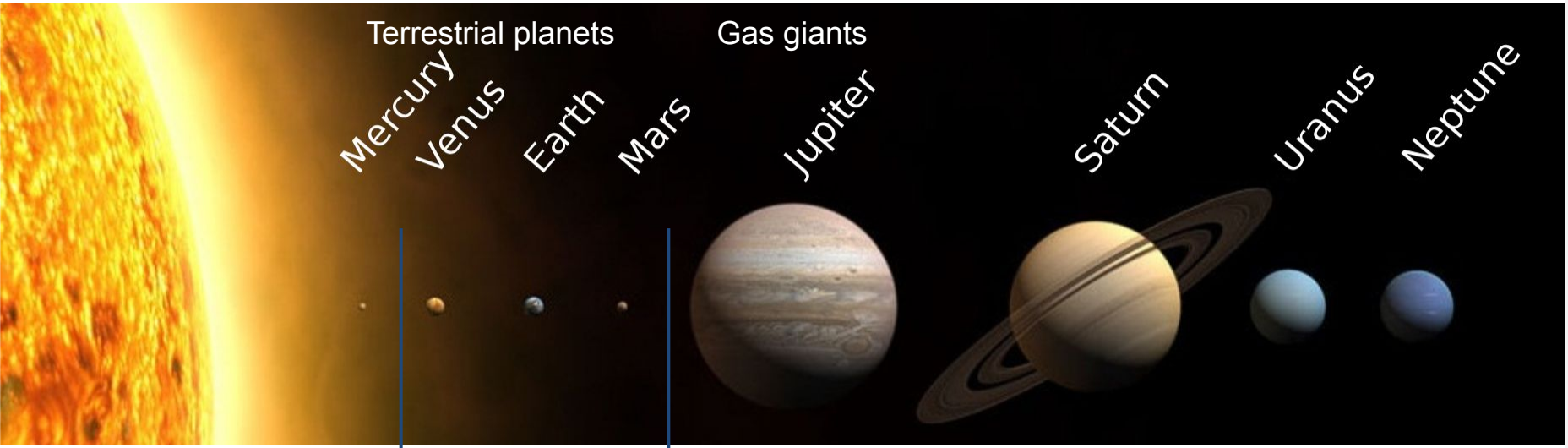
→ Planets with weaker gravity have a thinner atmosphere

→ Hotter planets would have a thinner atmosphere than colder ones



https://solarsystem.nasa.gov/planets/overview/

What is your guess about the atmospheres of these planets?



Both, light and heavy gas gets lost	Light gases get lost	Everything can be retained
Solid surface	Solid surface	No solid surface ..

Comparison

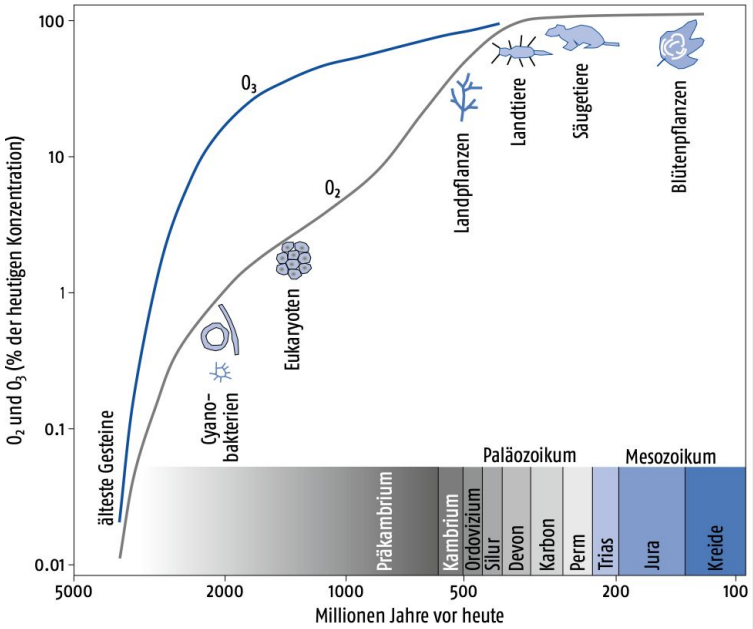


	Venus	Earth	Mars
Atmospheric pressure	90 bar	1 bar	0.007 bar
Composition	CO_2 (96%) N_2 (3.5%)	N_2 (78%) O_2 (21%) Ar (1%) CO_2 (0.04%) H_20 (variabel)	CO_2 (95%) N_2 (2.7%)
Surface temperature	470 °C	15 °C	-50 °C

Evolution of the Earth's atmosphere

Atmospheres change over the period of their planets' evolution

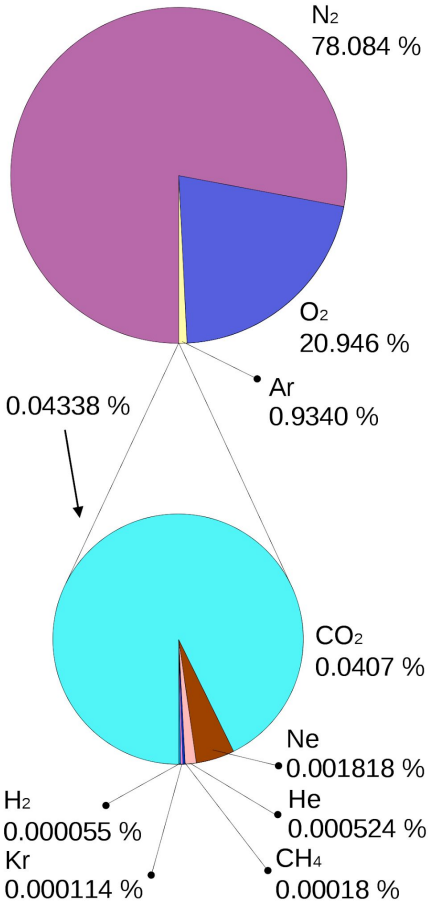
- Earth initially hydrogen, helium, methane, ammonia → losses because "too light" and "too fast".
- Then water, CO_2, H_2S, … Slow increase of O_2 by "life" after complete oxidation, later formation of an ozone layer
- Accumulation of N_2



Taken from: Brönnimann (2018) *Klimatologie*, utb, Haupt Verlag, Berlin

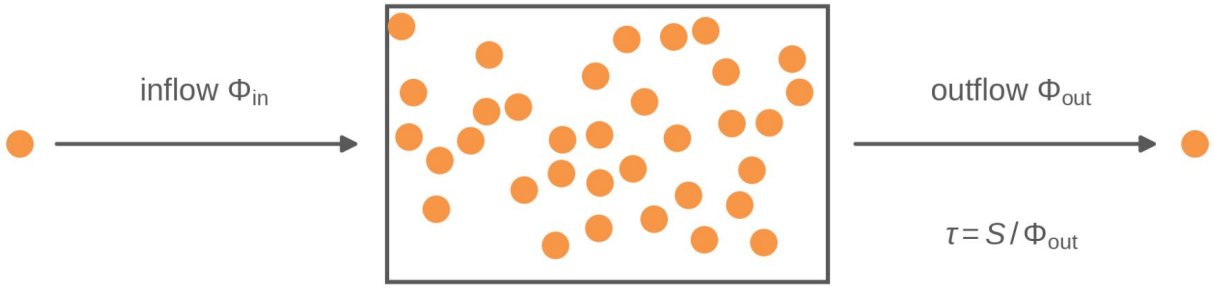
COMPOSITION OF OUR ATMOS

Composition of the (lower) atmosphere



Wikicommons, Nov. 2020

As with any reservoir, gases have a characteristic turnover



$$\tau = \frac{S}{\Phi_{\text{out}}} = \frac{S}{\Phi_{\text{in}}}$$

τ	residence time (turnover time) [d] or [a]
S	stock, e.g. water in the atmosphere [Gt]
Φ_{in}, Φ_{out}	inflow, outflow [$\text{Gt}\,\text{d}^{-1}$]

Formula sheet I-4

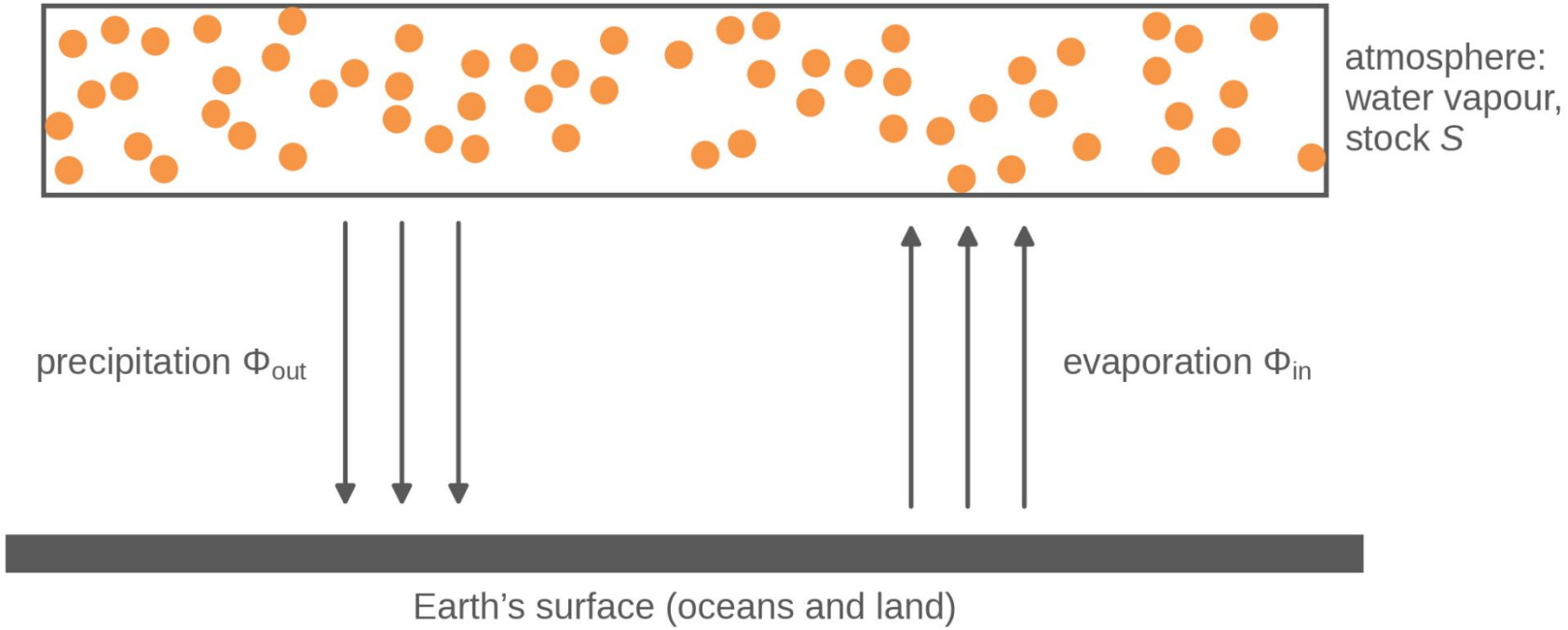
As with any reservoir, gases have a characteristic turnover

$$\tau = \frac{S}{\Phi_{out}} = \frac{S}{\Phi_{in}}$$

τ	residence time (turnover time) [d] or [a]
S	stock, e.g. water in the atmosphere [Gt]
Φ_{in}, Φ_{out}	inflow, outflow [$Gt\,d^{-1}$]

Species	Lifetime (years) [1]:731
CO2 [ppm]	100 *
CH4 [ppb]	12.4
N2O [ppb]	121
CFC-12	100
CFC-11	45
HCFC-22	11.9
CFC-113	85
CCl4	26
HFC-134a	13.4
HFC-23	222
SF6	3200
HCFC-142b	17.2
CF4 (PFC-14)	50000
HCFC-141b	9.2
HFC-125	28.2
HFC-143a	47.1
CFC-115	1020
C2F6 (PFC-116)	10000
CFC-13	640
HFC-152a	1.5
HFC-32	5.2

How long will water last in the atmosphere?



Your turn: how long does water stay?

1. The atmosphere holds about 12.7 × 10^3 km³ of water vapour; global precipitation is about 486 × 10^3 km³ per year. **How long does a water molecule stay in the atmosphere on average? Give the answer in days.**

2. The ocean holds about 1.335 × 10^9 km³ of water; about 413 × 10^3 km³ evaporate from it per year. **What is the residence time of water in the ocean?**

3. Compare both results. **What does the difference mean for how quickly the atmosphere and the ocean respond to a change?**

4. Why do we need the assumption of a steady state (inflow = outflow)? **What happens to τ if the outflow increases?**

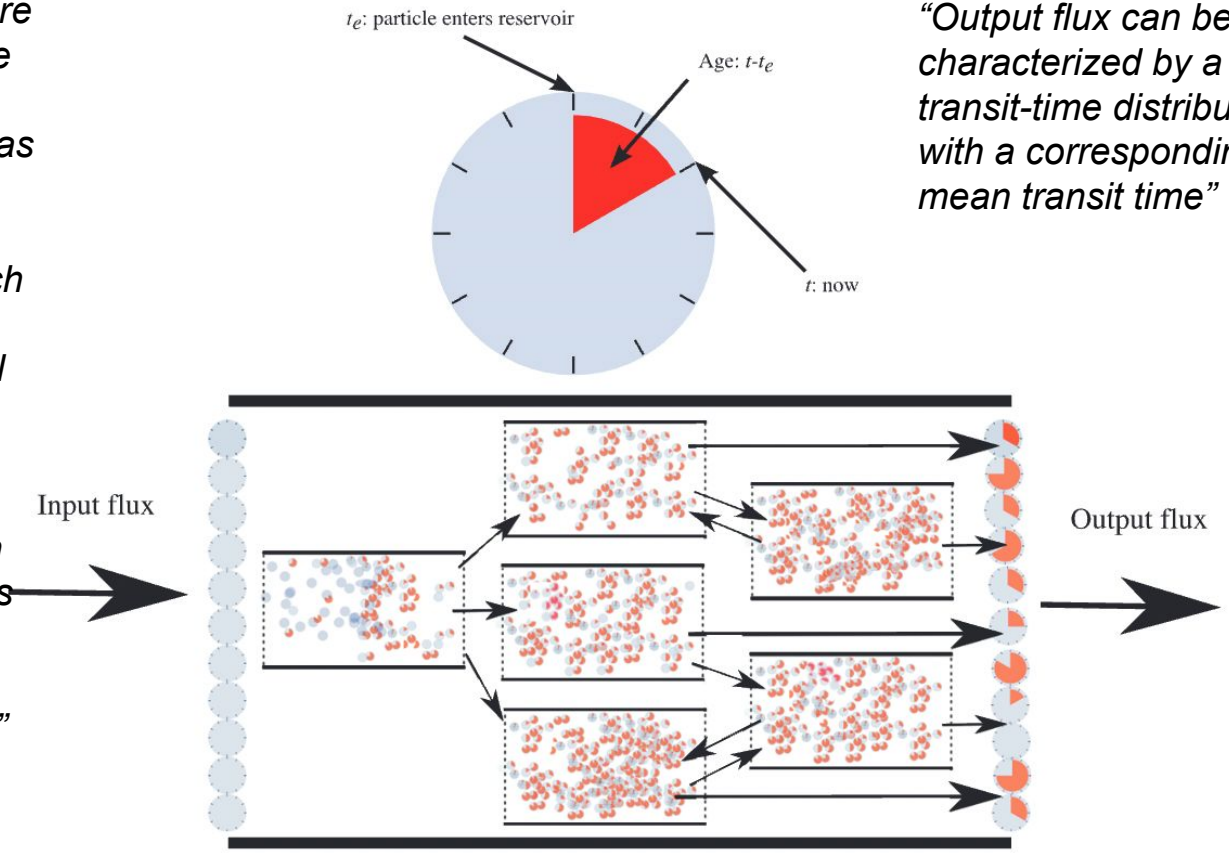
Values: Trenberth et al. (2007), J. Hydrometeorol. 8, 758–769. 1 km³ of water ≈ 1 Gt.

How you should really think about this

Sierra, C. A., et al (2017). *Global Change Biology*, *23*(5), 1763-1773

"Each molecule is represented here as a clock that measures their age in the system. A system (e.g. the atmosphere) can be represented as a set of pools (rectangles) with mass transfer among each other (arrows), with an input flux in which particles enter with age=0. At any given time t, particles in each pool have different ages and therefore each pool has a pool-age distribution with a corresponding mean pool age. For all particles in the system at any given time t, it is also possible to define a system-age distribution with a corresponding mean system age." Sierra et al. (2017)

(Text slightly adapted)



"Output flux can be characterized by a transit-time distributio with a corresponding mean transit time"